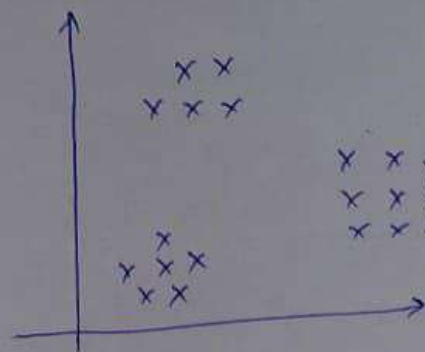
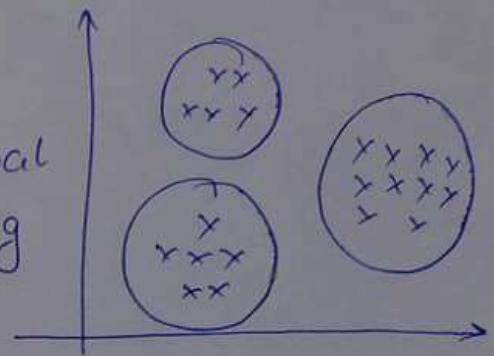




⇒ Hierarchical clustering

$K=3$ No Centroids

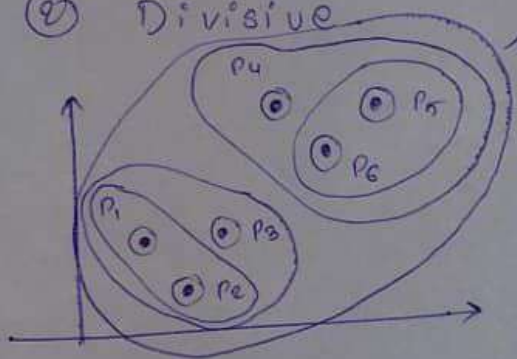


HC

① Agglomerative

② Divisive

⇒ Geometric Intuition



steps

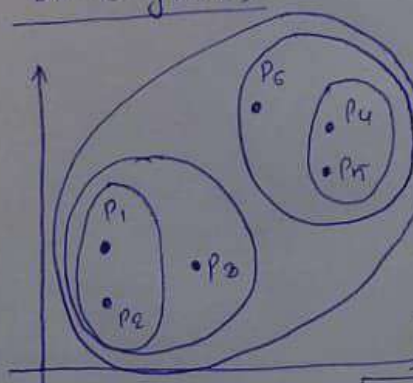
① For each point initially will consider it as a separate cluster.

② Find the nearest point and create a new cluster.

③ Keep on doing the same process until we get a single cluster

No. of clusters

Dendrogram



Eucledian Distance

Eucledian Distance

$K=2$

↑↑↑↑

Threshold

$K=4$

$K=6$

Dendrogram

$K=2$

$K=2$

$P_1, P_2, P_3, P_4, P_5, P_6$

P_1, P_2, P_3

P_1, P_2

P_4, P_5

P_4, P_5, P_6

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2

P_4, P_5

P_4, P_5, P_6

P_1, P_2, P_3

P_1, P_2, P_3

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P_4, P_5, P_6

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P_4, P_5, P_6

P_1, P_2, P_3

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P_1, P_2

P_4, P_5

P_4, P_5, P_6

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2

P_4, P_5

P_4, P_5, P_6

P_1, P_2, P_3

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P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P_3

P_1, P_2, P